WORKED SOLUTION

STEP II 1996, Question 7

Pure • Loci in Coordinate Geometry

STEP II 1996, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s2-q7

The key idea

Put the square in coordinates so that it's symmetric about both axes. Then each distance is the modulus of a linear expression, and the two products pr and qs become $|y^2 - a^2|$ and $|x^2 - a^2|$. An equation of the form $|U| = |V|$ splits into $U = V$ or $U = -V$. One case gives the two lines and the other gives the circle.

Solution

Let the square have side $2a$ (with $a > 0$) and centre at the origin, with sides parallel to the axes:

$$A = (-a, -a), \quad B = (a, -a), \quad C = (a, a), \quad D = (-a, a).$$

Then AB lies on the line $y = -a$, BC on $x = a$, CD on $y = a$ and DA on $x = -a$. The perpendicular distance from $P = (x, y)$ to a horizontal line $y = c$ is $|y - c|$, and to a vertical line $x = c$ it is $|x - c|$. So

$$p = |y + a|, \qquad q = |x - a|, \qquad r = |y - a|, \qquad s = |x + a|.$$

These formulae hold for every P in the plane, inside or outside the square. The moduli are what make the distances non-negative, so we mustn't drop them.

Multiplying in pairs, and using $|u|\,|v| = |uv|$,

$$pr = |(y + a)(y - a)| = |y^2 - a^2|, \qquad qs = |(x - a)(x + a)| = |x^2 - a^2|.$$

Ask yourself How do we remove the moduli from $|y^2 - a^2| = |x^2 - a^2|$ without splitting into lots of regions?

For real numbers, $|U| = |V|$ holds if and only if $U = V$ or $U = -V$, because both sides are non-negative and $|U| = |V| \iff U^2 = V^2 \iff (U - V)(U + V) = 0$. Since we only ever use "if and only if" steps, we get the necessary and the sufficient conditions

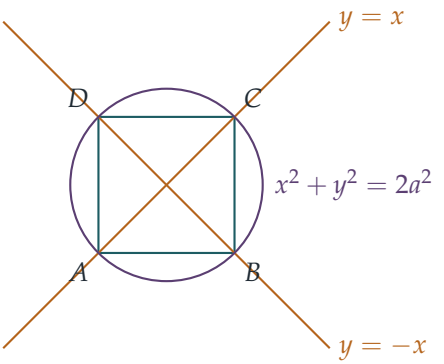
at the same time. So $pr = qs$ holds if and only if

$$y^2 - a^2 = x^2 - a^2 \qquad \text{or} \qquad y^2 - a^2 = -(x^2 - a^2).$$

First case. $y^2 = x^2$, that is, $(y - x)(y + x) = 0$, so $y = x$ or $y = -x$. These are the two (infinite) lines through the centre of the square that contain the diagonals AC and BD.

Second case. $x^2 + y^2 = 2a^2$. This is the circle with centre the centre of the square and radius $\sqrt{2}\,a$, which is half the diagonal. So it is the circumcircle of the square, passing through A, B, C and D.

Hence $pr = qs$ forces P to lie on one of the two diagonal lines AC, BD (extended indefinitely) or on the circumcircle of $ABCD$. Conversely, every step above was reversible. If P is on either line, then $x^2 = y^2$, so $|y^2 - a^2| = |x^2 - a^2|$. If P is on the circle, then $y^2 - a^2 = a^2 - x^2$, which again gives $|y^2 - a^2| = |x^2 - a^2|$. Either way $pr = qs$, so every point of the two lines and the circle is a possible position of P. This completes the proof.



Ask yourself Is a point outside the square really allowed, and do the lines stop at the vertices?

The question allows any point in the plane, and our distance formulae are valid everywhere. So the locus is the whole of each line, not just the diagonals as segments. For example, $P = (3a, 3a)$ has $p = 4a$, $r = 2a$, $q = 2a$, $s = 4a$, so $pr = qs = 8a^2$.

Checks

- The centre $(0, 0)$ is on both lines: $p = q = r = s = a$, so $pr = qs$.
- Take the point of the circle directly below the midpoint of AB, namely $(0, -\sqrt{2}\,a)$. Then $p = (\sqrt{2} - 1)a$, $r = (\sqrt{2} + 1)a$, so $pr = a^2$, and $q = s = a$, so $qs = a^2$.
- A point not on the locus, such as $(\frac{1}{2}a, 0)$, gives $pr = a^2$ but $qs = \frac{3}{4}a^2$, so it fails.
- Numerically (with $a = 1.7$), $pr = qs$ held to 10^{-9} at 30 000 random points on the three curves. A sign-change scan of $pr - qs$ over a 600×600 grid on $[-5, 5]^2$ found no zeros away from the two lines and the circle.

Common slips

- Dropping the moduli and writing $pr = a^2 - y^2$, $qs = a^2 - x^2$. This only works inside the square and loses the circle, giving just $y = \pm x$.
- Proving only that P must lie on the curves. The question also asks for the converse, so say why every point of the lines and the circle works.
- Describing the lines as the diagonals *segments* AC and BD. They are full lines.

Beyond STEP

The locus is the zero set of the polynomial $(y^2 - a^2)^2 - (x^2 - a^2)^2$, which factorises as $(y - x)(y + x)(x^2 + y^2 - 2a^2)$. A curve given by a polynomial equation that splits into simpler pieces like this is a first example of a reducible *algebraic curve*. Choosing coordinates to match a figure's symmetry is the habit that makes such factorisations visible. For more geometry in this style, see Siklos, *Advanced Problems in Mathematics*, and for the algebra of curves, Reid, *Undergraduate Algebraic Geometry*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.